

MD2G-Cast Compact Figure Preview

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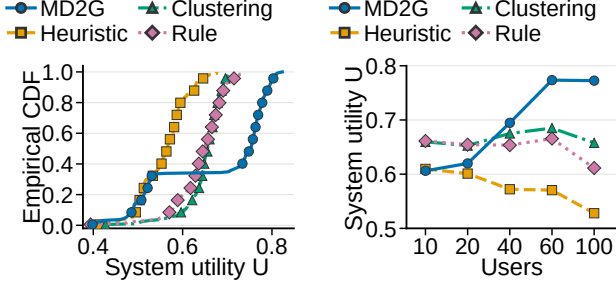


Figure 1: Utility ECDF over development workloads.

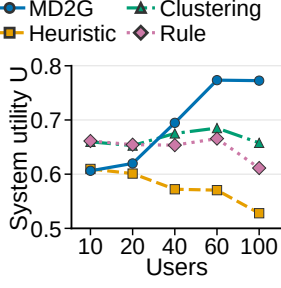


Figure 2: Utility versus re-development workloads.

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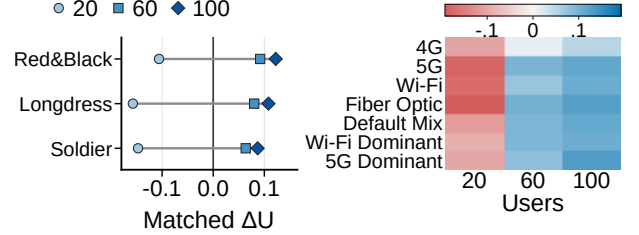


Figure 3: Content-wise matched ΔU across loads.

1 EVALUATION

We evaluate MD2G under the frozen nested-component contract on development workloads and a sealed Loot hold-out. Same-substrate comparisons use Heuristic, Clustering, and Rule. Canonical utility is $U = \text{clip}(0.25R_o + 0.60R_q - 0.15R_b, 0, 1)$. Figures 1–8 are independent floats placed in single-column pairs for layout only.

On MAINDEV, mean matched ΔU versus the strongest same-substrate baseline is +0.016, with MD2G mean $U \approx 0.675$. The sign of the matched gap depends on load: negative at 20 users and positive at 60 and 100 users. Scaling cells at $u \in \{10, 20, 40, 60, 100\}$ reproduce the same crossover. These facts are supporting context for the compact figures below; they do not change frozen metrics.

Fig. 1 shows the MAINDEV utility distribution. Fig. 2 reports mean U at $u \in \{10, 20, 40, 60, 100\}$. The same-substrate gain appears at medium and high concurrency; low concurrency remains a limitation. We do not interpret this crossover as universal dominance, and we do not retune after holdout.

Robustness over content and access is reported next. Each matched block keeps all launched users in the denominator. Display labels Red&Black, Longdress, and Soldier are short names for the same frozen contents. Network rows keep all seven access profiles, including constrained 4G.

Matched ΔU in Figs. 3 and 4 is MD2G minus the strongest same-substrate baseline. Negative cells at 20 users are retained. The heatmap uses a diverging scale with no in-cell numerals so the load-conditioned sign change remains the primary visual message.

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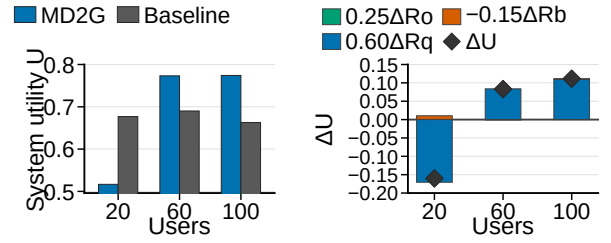


Figure 4: Network-wise matched ΔU across loads.

Loot remained network-sealed until FINAL DEV freeze and is evaluated as a 315-cell holdout under the same epoch and controller. The holdout pair below is the load-conditioned story, not an aggregate-only claim.

On Loot (Fig. 5), MD2G trails at 20 users ($\Delta U \approx -0.160$) and leads at 60 and 100 users ($\Delta U \approx +0.083$ and $+0.111$). Fig. 6 attributes the signed ΔU to weighted R_o , R_q , and R_b terms; it is a decomposition of the observed utility gap, not an ablation.

The last pair separates architecture from cross-stack systems. Shared MoQ versus MoQ unicast is not a DASH comparison. GROOT and Rolling remain authentic DASH/HTTP unicast baselines and are plotted only in Fig. 8.

Fig. 7 isolates delivery architecture (shared MoQ versus MoQ unicast, $R_o=0$), not a controller ranking. Fig. 8 is a separate cross-stack DASH/HTTP comparison and must not be read as MoQ unicast.

